

DSAINTE Demonstrator Node 01 Masterplan

Distributed Sovereign AI Infrastructure Nodes Technology

Demonstrator Node 01 – Bedburg-Hau, Germany

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Project Status: Demonstrator Node Masterplan

Disclaimer: Concept Development Document – not an offer of securities or investment solicitation.

Executive Summary

The DSAINT Demonstrator Node 01 Masterplan defines the first operational implementation of the DSAINT distributed sovereign AI infrastructure network.

Located in Bedburg-Hau, North Rhine-Westphalia, Germany, Demonstrator Node 01 serves as the reference design for future DSAINT infrastructure nodes across Germany and Europe.

The facility is designed as a Tier III AI infrastructure campus supporting artificial intelligence, high-performance computing, sovereign cloud services, research computing, and future digital workloads.

Initial capacity is planned at 2 MW IT Load with future expansion capability up to 6 MW.

Project Objectives

Technical Objectives

- Demonstrate DSAINT infrastructure architecture
- Validate distributed node concept
- Support AI and HPC workloads
- Achieve Tier III operational standards
- Maintain target PUE ≤ 1.20

Strategic Objectives

- Establish DSAINT's first operational node
- Create a replicable infrastructure template
- Support regional innovation ecosystems
- Strengthen digital sovereignty

Economic Objectives

- Generate long-term infrastructure revenues
 - Create regional employment opportunities
 - Attract technology investment
 - Establish strategic partnerships
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Site Overview

Location

Bedburg-Hau

North Rhine-Westphalia

Germany

Site Function

Regional AI Infrastructure Campus

Infrastructure Classification

Critical Digital Infrastructure

Design Life

30+ Years

Expansion Capability

Designed for phased capacity growth.

Approved Site Layout



1. AI Infrastructure Core

Primary facility for:

- AI Training
- AI Inference
- HPC Workloads
- Sovereign Cloud Services

Specifications:

- Tier III design
- Direct-to-chip liquid cooling
- Modular expansion capability

2. Admin & Operations Building

Specifications:

- 3 floors

- DSAINT branding on front façade
- Rooftop helipad

Functions:

- Operations Center
 - Security Operations Center
 - Management Offices
 - Meeting Rooms
 - Training Facilities
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3. Maintenance & Storage Building

Specifications:

- 2 floors

Ground Floor:

- Storage
- Logistics
- Spare Parts

Upper Floor:

- Workshops
 - Maintenance Operations
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4. Dedicated Substation

Specifications:

20kV / 110kV

Purpose:

- Grid connection
- Reliable power delivery
- Future expansion support

Location:

Upper-right sector of the site.

5. Technical Cooling Plant

Components:

- Dry coolers
- Heat exchangers
- Pump systems
- Thermal buffers
- Water treatment systems

Purpose:

- Cooling infrastructure
 - Energy efficiency
 - System redundancy
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6. Parking Area

Purpose:

- Employee parking
- Visitor parking
- Service access

Additional Function:

BTES geothermal field located beneath parking and adjacent infrastructure areas.

7. Security Airlock

Purpose:

Controlled access to critical infrastructure zones.

Functions:

- Personnel verification
 - Vehicle screening
 - Security monitoring
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8. Main Entrance

Connection:

Udemer Straße (L177)

Functions:

- Employee access
 - Visitor access
 - Administrative entry
-

9. Service Entrance

Connection:

Johann-van-Aken-Ring

Functions:

- Logistics
 - Deliveries
 - Maintenance vehicles
-

10. Employee Housing Building

Specifications:

- 5 floors

Functions:

- Staff accommodation
 - Emergency operations support
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11. Children's Playground

Purpose:

Support employee well-being and family-friendly workplace environment.

12. Guest Center

Purpose:

- Municipal meetings
- Investor visits
- Technology demonstrations

- Public engagement

Functions:

- Conference facilities
 - Presentation rooms
 - Visitor reception
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Capacity Plan

Initial Deployment

IT Capacity:

2 MW

Expansion Capacity

Maximum Planned Capacity:

6 MW

Rack Density

Designed for:

60–120 kW racks

Supported Workloads

- AI Training
 - AI Inference
 - HPC
 - Research Computing
 - Enterprise Computing
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Technical Infrastructure

Tier Classification

Tier III

Availability Target

99.982%

Redundancy

N+1

Cooling Technology

Direct-to-Chip Liquid Cooling

Benefits:

- High rack density
 - Improved efficiency
 - Future AI readiness
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Network Infrastructure

- Carrier-neutral design
 - Multi-fiber connectivity
 - High-bandwidth backbone
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Sustainability Masterplan

BTES System

Borehole Thermal Energy Storage

Location:

- Parking areas
- Internal roads

- Landscaped areas

Configuration:

- 12–20 boreholes
- 120–180 meters depth

Functions:

- Thermal balancing
 - Seasonal energy storage
 - Cooling assistance
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Heat Recovery

Future integration opportunities:

- District heating
 - Healthcare facilities
 - Agricultural applications
 - Municipal infrastructure
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Environmental Targets

Target PUE:

≤ 1.20

Long-term emphasis on:

- Energy efficiency
 - Sustainable cooling
 - Carbon reduction
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Security Masterplan

Perimeter 1

Site Security Zone

Includes:

- Fencing
- Vehicle control

- CCTV
 - Intrusion detection
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Perimeter 2

Critical Infrastructure Zone

Includes:

- Airlocks
 - Access control
 - Security monitoring
 - Restricted access areas
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Security Operations Center

Located within Admin & Operations Building.

24/7 monitoring capability.

Operational Model

Infrastructure Ownership

DSAINTE owns:

- Land
 - Buildings
 - Utilities
 - Power systems
 - Cooling systems
 - Security systems
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Customer Operations

Customers provide:

- GPUs
- AI accelerators
- Servers

- Applications
 - Workloads
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Financial Summary

Estimated CAPEX:

€43 Million

Estimated Annual OPEX:

€6.5 Million

Operational Model:

Infrastructure-as-a-Platform (IaaS)

Project Timeline

Phase 1

Concept Development

2026–2027

Phase 2

Engineering & Permitting

2027–2028

Phase 3

Financing & Procurement

2028–2029

Phase 4

Construction

2029–2030

Phase 5

Commissioning

2030

Strategic Importance

Demonstrator Node 01 serves as:

- The first DSAINT infrastructure node
 - Technical proof of concept
 - Operational benchmark
 - Replication template
 - Foundation of the future DSAINT network
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Long-Term Vision

Demonstrator Node 01 establishes the foundation for a future European network of distributed sovereign AI infrastructure nodes supporting digital sovereignty, infrastructure resilience, research, innovation, and sustainable economic development.

DSAIN

Distributed Sovereign AI Infrastructure Nodes Technology

Demonstrator Node 01 Masterplan

Bedburg-Hau, Germany

Building the Infrastructure of Europe's AI Future.